Riya Shet

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Profile

Data scientist and first-author MICCAI 2026 researcher. I build and deploy computer-vision and LLM systems in Python and PyTorch, from deep-learning segmentation models to RAG tools, and design the evaluation that shows when they can be trusted.

Publication and Research

Shet R.D., Zhang L. (2026). Reliability analysis for BraTS-GoAT segmentation: a controlled robustness study of deep-ensemble uncertainty. *MICCAI 2026 Satellite Events*, Springer LNCS 17254.

- Trained a 5-fold nnU-Net and a 3-seed deep ensemble; **whole-tumour Dice 0.87** on the official validation leaderboard; containerised (Docker) submission to the challenge evaluation.
- Under acquisition shift, single-model confidence stayed flat while accuracy fell; ensemble disagreement rose 25–33%, so it detected the shift that confidence missed.

Selected Projects

- **TracHeal, team lead (Harvard HSIL Hackathon, Apr 2026):** led four people to ship a live LLM-based retrieval-augmented (RAG) clinical decision-support tool in 48 hours; owned deployment.
- **Diabetic retinopathy computer-vision model audit:** a 0.88 headline kappa masked 48.6% sensitivity on the most urgent grade (PyTorch).
- **Market model with calibrated uncertainty:** LightGBM on 5.2M order-book records, correlation 0.17 with actual moves on all 97 test days; acting only on its most confident half cut error by 30%; six-page Power BI report.
- **AI triage simulation:** 10,000-iteration Monte Carlo on a MIMIC-IV cohort extracted with SQL in BigQuery.

Education

MSc Health Data Science, University of Birmingham Sep 2025 – Nov 2026
Dubai, UAE

Distinction (final average 77.35%). Modules: data analytics and statistical machine learning (79.8), integrative multimodal data analytics (78.2). Course Student Representative.

BSc (Hons) Psychology, CHRIST (Deemed to be University) 2019 – 2022
Bengaluru, India

First Class, 3.55/4.0; coursework in AI and neural networks. GATE 2025 qualified (All-India Rank 917).

Experience

Research Associate, SyNC Positive Psychiatry Apr – Jul 2025
Bengaluru, India

Built a KPI-based investment case and the official service-integration proposal for an R&D pilot.

Clinical Psychology Intern, DIMHANS Sep – Oct 2023
Dharwad, India

Psychometric assessment, high-volume case documentation, cross-department coordination.

Skills

Programming: Python (PyTorch, scikit-learn, pandas, SHAP), R, SQL (Google BigQuery)
Engineering: Docker, Linux HPC (BlueBEAR), Git/GitHub, Power BI; RAG applications
Agentic systems: built a five-agent review pipeline (orchestration with defined inputs and outputs, a rubric-based blind grader) plus persistent context rules; it caught eleven errors before submission
Languages: English (IELTS 8.0), Hindi (fluent), Arabic (reads and writes, 12 years of school study; spoken basic)